

```

1  import Mathlib
2
3  /-!
4  # 🍷 Recipe for Exotic Matter (Casimir-vacuum edition)
5
6  **Exotic matter** (in general relativity) is matter that violates the *null energy
7  condition* (NEC): along some light ray,  $\rho + p < 0$ . Traversable wormholes and warp
8  drives need it, and Nature serves a tiny portion in the **Casimir effect**: between two
9  ideal parallel plates a distance `a` apart, the vacuum has energy density
10
11      $\rho = -\pi^2 \hbar c / (720 a^4) < 0$ .
12
13  Below: pantry → ingredients → method → serve. Lean *proves* the dish is exotic.
14
15  Run it at https://live.lean-lang.org (the "Mathlib" project) – nothing to install.
16  -/
17
18  noncomputable section
19
20  namespace ExoticMatter
21
22  /-! ## 1. The pantry: stress-energy and the energy conditions -/
23
24  /-- Static diagonal stress-energy tensor in an orthonormal frame:
25  energy density `rho` and principal pressures `px py pz` (SI units, per m³). -/
26  structure StressEnergy where
27      rho : ℝ
28      px  : ℝ
29      py  : ℝ
30      pz  : ℝ
31
32  /-- Null energy condition:  $\rho + p_i \geq 0$  in every principal direction. -/
33  def NEC (T : StressEnergy) : Prop :=
34      0 ≤ T.rho + T.px ∧ 0 ≤ T.rho + T.py ∧ 0 ≤ T.rho + T.pz
35
36  /-- Weak energy condition: non-negative energy density, plus the NEC. -/
37  def WEC (T : StressEnergy) : Prop := 0 ≤ T.rho ∧ NEC T
38
39  /-- Strong energy condition: the NEC, plus  $\rho + px + py + pz \geq 0$ . -/
40  def SEC (T : StressEnergy) : Prop := NEC T ∧ 0 ≤ T.rho + T.px + T.py + T.pz
41
42  /-- Dominant energy condition: energy density dominates every pressure,  $\rho \geq |p_i|$ . -/
43  def DEC (T : StressEnergy) : Prop := |T.px| ≤ T.rho ∧ |T.py| ≤ T.rho ∧ |T.pz| ≤ T.rho

```

```

44
45 /-- **Exotic matter**: stress-energy that violates the null energy condition. -/
46 def Exotic (T : StressEnergy) : Prop := ¬ NEC T
47
48 /-- A perfect fluid: energy density `rho`, isotropic pressure `p`. -/
49 def fluid (rho p : ℝ) : StressEnergy := ⟨rho, p, p, p⟩
50
51 /-- Sanity check: ordinary matter (dust `p = 0`, radiation `p = ρ/3`, stiff matter
52 `p = ρ`, ...) is not exotic. -/
53 theorem ordinary_not_exotic {rho p : ℝ} (h₀ : 0 ≤ p) (h₁ : p ≤ rho) :
54   ¬ Exotic (fluid rho p) := by
55     intro h
56     have hsum : 0 ≤ rho + p := by linarith
57     exact h ⟨hsum, hsum, hsum⟩
58
59 /-- Plain empty vacuum: nothing in, nothing out. -/
60 def plainVacuum : StressEnergy := fluid 0 0
61
62 theorem plainVacuum_not_exotic : ¬ Exotic plainVacuum :=
63   ordinary_not_exotic (rho := 0) (p := 0) le_rfl le_rfl
64
65 /-! ## 2. The ingredients -/
66
67 /-- Everything you need from the cupboard. -/
68 structure Ingredients where
69   hbar : ℝ -- reduced Planck constant, J·s
70   c    : ℝ -- speed of light, m/s
71   gap  : ℝ -- distance between the plates, m
72   hbar_pos : 0 < hbar
73   c_pos   : 0 < c
74   gap_pos : 0 < gap
75
76 /-- A well-stocked lab: real SI constants, plates one micrometre apart. -/
77 def pantry : Ingredients where
78   hbar := 1054571817 / 10 ^ 43 -- 1.054571817 × 10-34 J·s
79   c    := 299792458             -- m/s
80   gap  := 1 / 10 ^ 6            -- 1 μm
81   hbar_pos := by norm_num
82   c_pos   := by norm_num
83   gap_pos := by norm_num
84
85 /-! ## 3. The dish: the Casimir vacuum -/
86

```

```

87 /-- Casimir energy density between ideal parallel plates:  $\rho = -\pi^2 \hbar c / (720 a^4)$ . -/
88 def casimirRho (I : Ingredients) : ℝ :=
89   -(Real.pi ^ 2 * I.hbar * I.c / (720 * I.gap ^ 4))
90
91 /-- Casimir stress-energy  $T = \rho \cdot \text{diag}(1, -1, -1, 3)$  (traceless;  $z \perp$  plates). -/
92 def casimirTensor (I : Ingredients) : StressEnergy where
93   rho := casimirRho I
94   px  := -casimirRho I
95   py  := -casimirRho I
96   pz  := 3 * casimirRho I
97
98 /-- The Casimir vacuum has negative energy density. -/
99 theorem casimirRho_neg (I : Ingredients) : casimirRho I < 0 := by
100   have h1 := I.hbar_pos
101   have h2 := I.c_pos
102   have h3 := I.gap_pos
103   have h : 0 < Real.pi ^ 2 * I.hbar * I.c / (720 * I.gap ^ 4) := by positivity
104   unfold casimirRho
105   linarith
106
107 /-- The Casimir vacuum violates the NEC (along  $z$ ): it is exotic matter. -/
108 theorem casimir_exotic (I : Ingredients) : Exotic (casimirTensor I) := by
109   intro h
110   obtain ⟨_, _, hz⟩ := h
111   have hz' : 0 ≤ casimirRho I + 3 * casimirRho I := hz
112   have hnec := casimirRho_neg I
113   linarith
114
115 /-- It is just as bad for the other energy conditions. -/
116 theorem casimir_not_wec (I : Ingredients) : ¬ WEC (casimirTensor I) :=
117   fun h => casimir_exotic I h.2
118
119 theorem casimir_not_sec (I : Ingredients) : ¬ SEC (casimirTensor I) :=
120   fun h => casimir_exotic I h.1
121
122 theorem casimir_not_dec (I : Ingredients) : ¬ DEC (casimirTensor I) := by
123   intro h
124   obtain ⟨h1, _, _⟩ := h
125   have hnec : (casimirTensor I).rho < 0 := casimirRho_neg I
126   have habs := abs_nonneg (casimirTensor I).px
127   linarith
128
129 /-! ## 4. The method -/

```

```

130
131 /-- State of the laboratory. -/
132 structure Kitchen where
133   evacuated : Bool
134   cold      : Bool
135   plates    : Bool
136   deriving Repr
137
138 /-- The three steps. Order matters! -/
139 inductive Step
140   | pumpDown      -- evacuate the chamber
141   | coolToGround  -- cryo-cool towards  $T \rightarrow 0$  (only works once evacuated)
142   | insertPlates  -- two parallel ideal conductors (only if evacuated *and* cold)
143
144 def Step.run : Step → Kitchen → Kitchen
145   | .pumpDown,      k => { k with evacuated := true }
146   | .coolToGround, k => { k with cold := k.evacuated }
147   | .insertPlates, k => { k with plates := k.evacuated && k.cold }
148
149 /-- Follow the steps, starting from an empty lab. -/
150 def bake (steps : List Step) : Kitchen :=
151   steps.foldl (fun k s => s.run k) ⟨false, false, false⟩
152
153 /-- The lab is ready when every condition holds. -/
154 def Kitchen.ready (k : Kitchen) : Bool := k.evacuated && k.cold && k.plates
155
156 /-- 🍳 The recipe. -/
157 def recipe : List Step := [.pumpDown, .coolToGround, .insertPlates]
158
159 /-- The oven: a ready lab yields the Casimir vacuum, anything else yields plain vacuum.
160 -/
161 def serve (steps : List Step) (I : Ingredients) : StressEnergy :=
162   if (bake steps).ready then casimirTensor I else plainVacuum
163
164 theorem recipe_ready : (bake recipe).ready = true := by decide
165
166 theorem serve_recipe (I : Ingredients) : serve recipe I = casimirTensor I := by
167   simp [serve, recipe_ready]
168
169 /-- 🍷 Bon appétit: following the recipe yields exotic matter, with any ingredients. -/
170 theorem recipe_yields_exotic_matter (I : Ingredients) : Exotic (serve recipe I) := by
171   rw [serve_recipe]
172   exact casimir_exotic I

```

```

172
173 /-- No shortcuts: insert the plates before pumping down and you just get plain vacuum.
174 -/
175 theorem wrong_order_gives_vacuum (I : Ingredients) :
176   serve [.insertPlates, .pumpDown, .coolToGround] I = plainVacuum := by
177     have h : (bake [Step.insertPlates, Step.pumpDown, Step.coolToGround]).ready = false :=
178       by
179         decide
180         simp [serve, h]
181
182 /-! ## 5. Serving suggestion: hold a wormhole open -/
183
184 /-- For a Morris-Thorne throat of radius `r_0` with shape function `b` (units `G = c = 1`),
185 the Einstein equations give, at the throat, ` $\rho = b'(r_0)/(8\pi r_0^2)$ ` and ` $p_r = -1/(8\pi r_0^2)$ `
186 `.`
187 The *flare-out* condition ` $b'(r_0) < 1$ ` then forces ` $\rho + p_r < 0$ `: NEC violation, i.e. ex
188 otic
189 matter. (Point the Casimir plates' normal along the radial direction.) -/
190 theorem throat_needs_exotic_matter {r_0 b' : ℝ} (hr : 0 < r_0) (hflare : b' < 1) :
191   b' / (8 * Real.pi * r_0 ^ 2) + (-1) / (8 * Real.pi * r_0 ^ 2) < 0 := by
192     have hD : 0 < 8 * Real.pi * r_0 ^ 2 := by positivity
193     have h : b' / (8 * Real.pi * r_0 ^ 2) + (-1) / (8 * Real.pi * r_0 ^ 2)
194       = (b' - 1) / (8 * Real.pi * r_0 ^ 2) := by ring
195     rw [h]
196     exact div_neg_of_neg_of_pos (by linarith) hD
197
198 /-! ## 6. Reality check
199
200 With a 1 μm gap the Casimir vacuum is only about -433 μJ/m³. Holding open a 1 m wormhol
201 e
202 throat (`b' = 0`) would need ` $\rho + p_r = -c^4 / (8\pi G r_0^2) \approx -4.8 \times 10^{42} \text{ J/m}^3$ `, roughly
203 45 orders of magnitude more. Great recipe for a physics lab; not yet for a starship. 🚀
204 -/
205
206 /-- Casimir yield in μJ/m³ (plain `Float`, so `#eval` can run it); `gap` in metres. -/
207 def casimirYield (gap : Float) : Float :=
208   1.0e6 * (-(3.141592653589793 * 3.141592653589793 * 1.054571817e-34 * 299792458.0)
209     / (720.0 * gap * gap * gap * gap))
210
211 #eval casimirYield 1.0e-6 -- plates 1 μm apart: about -433.375257 (μJ/m³)
212 #eval bake recipe -- { evacuated := true, cold := true, plates := true }
213 #print axioms recipe_yields_exotic_matter -- [propext, Classical.choice, Quot.sound]

```

```
208
209   end ExoticMatter
210
211   end
212
```